

Homework 3

This homework is due on Wednesday, October 22 2025 by 11:59pm. Please show all your work and justify your answers. Use R to compute the probabilities.

1. Trees are subjected to different levels of carbon dioxide atmosphere with 6% of them in a minimal growth condition at 350 parts per million (ppm), 10% at 450 ppm (slow growth), 47% at 550 ppm (moderate growth), and 37% at 650 ppm (rapid growth). What are the mean and standard deviation of the carbon dioxide atmosphere (in ppm) for these trees in ppm?
2. Suppose that in the U.S. in 2011, 70% of new start-ups failed in their first year. If there were 50 start-ups in the DMV area in 2011, what is the probability that 35 or more failed? What assumption did you need to make about the DMV?
3. A research team has developed a face recognition device to match photos in a database. From laboratory tests, the recognition accuracy is 92% and trials are assumed to be independent.
 - If the research team continues to run laboratory tests, what is the mean number of trials until failure?
 - What is the probability that the first failure occurs on the tenth trial?
 - To improve the recognition algorithm, a chief engineer decides to collect 10 failures. How many trials are expected to be needed?
4. You roll a die until obtaining two consecutive numbers that match. What is the probability that it takes exactly x rolls to do this, for x an integer greater than 1?
5. The MIT-BIT arrhythmia database contains 48 heart signal recordings with 22 and 26 from female and male patients, respectively. For a classification task, an analyst randomly selects 36 records for predictive model training and keeps the other 12 records for testing the model performance.
 - What is the expected number of female patient recordings selected for training?
 - What is the probability that at least three male patient recordings are selected for training?
 - What is the probability that the same number of male and female patients are used for testing the model performance?
6. An engineer is participating in a research project on the title patterns of junk emails. The number of junk emails which arrive in an individual's account every hour follows a Poisson distribution with a mean of 1.2.
 - What is the expected number of junk emails that an individual receives in an 8 hour day?
 - What is the probability that an individual receives more than two junk emails for the next three hours?
 - What is the probability that an individual receives no junk email for two hours?
7. Consider a random variable X with probability density function $f(x) = \theta x^{\theta-1}$ for $0 < x < 1$, where $\theta > 0$ is the parameter that controls the pdf.
 - Show that $f(x)$ is a valid pdf
 - Plot the pdf for $\theta \in \{1/3, 1/2, 1, 2, 3\}$. Describe how the distribution changes for different values of θ .
 - Find a general expression for the expected value and variance of X .
 - Find a general expression for the cumulative distribution function (cdf)

8. Web crawlers need to estimate the frequency of changes to Web sites to maintain a current index for Web searches. Assume that the changes to a Web site follow a Poisson process with a mean of 6 days. Let a random variable X denote the time (in days) until the next change.
- What is the probability that the next change occurs in less than 4.5 days?
 - What is the probability that the time until the next change is greater 9.5 days?
 - What is the time of the next change that is exceeded with probability 90%?